Business Rules

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Owner	Raghdah Al-Gahdari, Business Analyst
Approved by	Aaron (Supervisor)
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Applies to	AI Finance Decision Support — Team 1, built for IBM

1. Purpose

This document defines the materiality thresholds, variance classification logic, and category-specific rules used by the Assistant when detecting, calculating, classifying, and explaining financial variances.

It translates the findings in synthetic-data-research.docx (Section 2: Financial Variance and Materiality Research) into concrete, testable rules the Assistant applies against the project's dataset, as defined in CSV_SQL_Data_Structure_Specification.docx and Financial_Calculation_Formulas.md.

This document does not cover data provenance or confidentiality (see synthetic-data-policy.md), and does not cover external economic or industry drivers used to explain a variance (see market-context.md). It also does not determine whether the synthetic data itself is realistic — it defines the business logic the Assistant applies to whatever data it is given.

2. Scope

This document applies to the seven financial categories defined in the project schema (CSV_SQL_Data_Structure_Specification.docx, Section 3):

Category	Category_Group
Revenue	Revenue
Cost of Goods Sold	COGS
Payroll Expense	Expense
Marketing Expense	Expense
Rent Expense	Expense
Utilities Expense	Expense
Other Expense	Expense

3. Assumed Business Scenario

The research (synthetic-data-research.docx, Section 2.3) establishes that materiality and variance behaviour differ by industry. This project therefore adopts one explicit, fixed scenario, so that the thresholds in Section 5 have a real business justification rather than being arbitrary:

Assumed scenario: a mid-sized clothing retailer, Melbourne, with Cost of Goods Sold as its largest cost driver, moderate and relatively stable gross margins, and discretionary categories (Marketing, Other Expense) that are expected to fluctuate with campaign timing rather than signal a problem.

This scenario was chosen because it fits the category set already defined in the schema (a Revenue line, a COGS line tied to Revenue, and five expense lines split between fixed and discretionary spend) without requiring any change to the dataset structure. All thresholds and rules below are justified against this scenario, not against any property of the synthetic-data generator.

4. Variance Definitions

Term	Meaning	Relevance to Project
Variance	Difference between actual and budgeted/expected value	Core calculation (Formula 1, Financial_Calculation_Formulas.md)
Favourable variance	Actual result is better than budget (e.g. lower expenses, higher revenue)	Used to classify the result
Unfavourable variance	Actual result is worse than budget (e.g. higher expenses, lower revenue)	Used to classify the result
Cost variance	Difference between actual cost and budgeted cost	Applies to COGS and all Expense-group categories
Revenue variance	Difference between actual revenue and budgeted revenue	Applies to the Revenue category

4.1 Favourable / Unfavourable Classification Rule

Category_Group	Actual > Budget	Actual < Budget
Revenue	Favourable	Unfavourable
COGS	Unfavourable	Favourable
Expense	Unfavourable	Favourable

This rule applies to every category within a group. It is a fixed rule — no judgement is required to apply it.

5. Materiality: A Two-Part Test

Per the research (Section 2.4), materiality is not size alone — a variance is material if it could reasonably influence a decision-maker's judgement. Under the assumed scenario (Section 3), the Assistant applies both parts of the test below. A variance is treated as material if either part is triggered.

5.1 Part 1 — Quantitative Test (Rule-Based)

Fixed and predictable categories are given tighter thresholds than variable or discretionary ones, consistent with the research finding that “different financial items can have different levels of normal variation” (Section 2.3):

Category	Predictability	Percentage Variance Threshold	Justification
Rent Expense	Fixed (contractual lease)	3%	Contractual costs should not move without a known cause; any deviation is worth surfacing
Payroll Expense	Fixed/semi-fixed	5%	Headcount-driven, changes gradually; a mid-sized clothing retailer would not expect large swings outside a hiring/restructuring event

Category	Predictability	Percentage Variance Threshold	Justification
Revenue	Demand-driven, core metric	8%	Directly affects overall financial performance (Research 2.3.1). The 0.5–1% materiality benchmark for total revenue is too tight for a single line item at operating level, so 8% is used as the operating-variance threshold, distinct from statement-level materiality
Cost of Goods Sold	Variable, tied to Revenue and supplier pricing	8%	Expected to move with sales volume and supplier costs; set at the same level as Revenue because COGS moves directly with sales
Utilities Expense	Semi-variable (seasonal/usage)	20%	Usage- and season-driven; normal fluctuation is expected and does not by itself indicate an issue
Marketing Expense	Discretionary	30%	Campaign-timing driven; deviations commonly reflect planned business activity, not a problem (Research 2.3.1)
Other Expense	Discretionary/miscellaneous	38%	Least predictable category by nature; a wide band avoids flagging routine, low-materiality fluctuations

Threshold Note: The thresholds presented above are management defaults developed for the assumed scenario (a mid-sized clothing retailer, Melbourne). Where applicable, thresholds were calibrated at approximately 2.5x the standard deviation of normal random variation (noise_std_pct) built into the dataset generator for each category, consistent with the statistical Empirical Rule (Wikipedia contributors, 2026) and the logic used in Statistical Process Control, where control limits are set at 2.5-3x standard deviation to distinguish genuine signals from normal random fluctuation. This ensures that a flagged variance is unlikely to result from random noise alone (less than ~1% probability) and more likely reflects an actual business change. These thresholds are intended to support consistent variance assessment within this project and should not be interpreted as universal accounting or audit materiality thresholds. They were reviewed against the dataset generation assumptions and approved by the Supervisor.

5.2 Part 2 — Qualitative Test (Judgement-Based)

A variance is also treated as material, regardless of size, if any of the following apply:

- It relates to a category with compliance or regulatory reporting relevance.
- It suggests a possible data or process error (e.g. a value flagged under the Edge Case Handling Rules in Financial_Calculation_Formulas.md).
- It could reasonably be expected to affect a decision a reader of the analysis would make.

The qualitative test cannot be reduced to a fixed rule — it requires human judgement. The Assistant's role here is limited to flagging the variance and the reason it may be qualitatively relevant; it does not decide materiality on the analyst's behalf.

5.3 Data Verification Prompt

When any calculated result is negative (for example a negative variance, growth rate or gross margin), the Assistant shows the result as usual, with its explanatory note and full source traceability, followed by this message: “This result is negative. Please check the source figures and the calculation manually to make sure the data is correct before taking any action.” The Assistant does not conclude that the data is wrong. The decision remains with the financial analyst.

6. Rule-Based vs. Judgement-Based Behaviour

The research asks explicitly which Assistant behaviours should be rule-based and which should involve judgement. This project draws that line as follows:

Behaviour	Basis	Owner of the Decision
Calculating Dollar Variance / Percentage Variance	Rule-based (fixed formula)	Assistant
Classifying Favourable / Unfavourable	Rule-based (Section 4.1)	Assistant
Comparing variance against the quantitative threshold (Section 5.1)	Rule-based	Assistant
Flagging a qualitative materiality trigger (Section 5.2)	Judgement-informed, but Assistant only flags — does not conclude	Assistant flags, Analyst confirms
Identifying a possible external driver (economic/industry/market)	Judgement-informed, using market-context.md, always in probabilistic language	Assistant proposes, Analyst evaluates
Deciding whether to investigate further or take action	Judgement	Financial Analyst only

The Assistant must never present a judgement-based output (Section 5.2, or a driver from market-context.md) with the same certainty as a rule-based calculation. Rule-based outputs are stated as facts; judgement-informed outputs are stated as possibilities requiring analyst review.

7. Assistant Workflow

Detect → Calculate → Classify → Explain → Present → Human Decision

Step	Assistant Action	Basis
Detect	Identify whether a variance exists for the requested category/period	Rule-based — Financial_Calculation_Formulas.md
Calculate	Compute the requested formula (Dollar/Percentage Variance, Gross Margin, PoP, YoY, Contribution)	Rule-based — Financial_Calculation_Formulas.md
Classify	Apply Section 4.1 (direction) and Section 5.1 (quantitative threshold)	Rule-based — this document
Explain	Flag any qualitative trigger (Section 5.2) and any relevant external driver (market-context.md), in probabilistic language	Judgement-informed — this document + market-context.md
Present	Present the result with full source traceability (file, row, column)	Rule-based — Financial_Calculation_Formulas.md, Source Traceability Requirements
Human decision	The financial analyst decides whether further investigation or action is required	Judgement — Analyst only

The Assistant must not send alerts, escalate cases, approve budgets, override figures, or take corrective action at any step.

8. Relationship to Other Governing Documents

Document	Relationship to this Document	Owner
synthetic-data-policy.md	Governs provenance and confidentiality of the underlying dataset; this document assumes that policy's requirements are already satisfied	Business Analyst
market-context.md	Supplies the external drivers used in the "Explain" step (Section 7)	Business Analyst
Financial_Calculation_Formulas.md	Defines the calculation logic this document's classification and threshold rules are applied on top of	Business Analyst
CSV_SQL_Data_Structure_Specification.docx	Defines the categories, fields, and schema this document's rules reference	Technical team, reviewed by BA

9. Governance

- Any change to the assumed scenario (Section 3) requires BA sign-off and a version increment.
- Any change to the threshold values (Section 5.1) must be reviewed by the BA and formally approved by the Supervisor before implementation.
- This document moves from "Proposed" to "Approve" once the scenario and thresholds are formally confirmed.

Changelog

Version	Date	Change	Author
0.1	2026-09-10	Initial draft, derived from synthetic-data-research.docx Section 2	Raghdah Al-Gahdari
1.0	2026-10-08	Status changed to Approved. Revenue justification corrected from 5% to 8% to match the threshold. Scenario updated to mid-sized clothing retailer, Melbourne, and aligned in Section 3, the Payroll and COGS justifications and the Threshold Note. Threshold Note updated to reflect approval.	Raghdah Al-Gahdari
1.1	2026-10-16	Added Section 5.3 (Data Verification Prompt): any negative result is shown with a message asking the financial analyst to verify the data before taking action.	Raghdah Al-Gahdari

10. References

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- What is Variance Threshold? Definition, Process & Key Metrics. (n.d.). https://www.hyperbots.com/glossary/variance-threshold#formula-for-calculating-variance
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